

Computer Networks Lab

Lab 4: Basic Router Access and CLI Configuration

1. Objective

- Understand how to connect networking devices to a computer for configuration.
- Learn about console ports and console cables.
- Understand GUI and CLI interfaces.
- Access router CLI in Cisco Packet Tracer.
- Learn basic Cisco IOS modes and commands.
- Understand configuration files and basic security settings.

2. Basic Theory

Network devices such as routers and switches must be configured before they can operate in a network. Configuration is usually done through the Command Line Interface (CLI).

There are two main ways to interact with networking devices:

- GUI (Graphical User Interface)
- CLI (Command Line Interface)

In networking, CLI is widely used because it provides more control and flexibility.

3. Connecting to Network Devices

To configure routers or switches directly, a computer must be connected to the device using a console connection.

Common connection components:

- Console Port on router/switch
- RJ45 Console Cable
- DB9 Connector (older PCs)
- USB Mini Console Port (newer devices)
- Rollover Cable used for console connection

4. Terminal Emulator Software

To communicate with the device through the console connection, terminal emulator software is used. Examples include:

- PuTTY
- SSH terminal programs

These tools allow the user to access the device CLI from the computer.

5. Cisco CLI Modes

Cisco devices operate using different CLI modes.

User EXEC Mode

Basic access mode for monitoring commands.

Prompt example: Router>

Privileged EXEC Mode

Provides advanced access.

Command: enable

Prompt example: Router#

Global Configuration Mode

Used to configure device settings.

Command: configure terminal

Prompt example: Router(config)#

6. Basic Commands

enable – Enter privileged mode.

configure terminal – Enter global configuration mode.

enable password – Set password for privileged mode.

7. Configuration Files

Running Configuration

- Active configuration stored in RAM.
- Command: show running-config

Startup Configuration

- Configuration stored in NVRAM.
- Command: show startup-config

8. Saving Configuration

Commands used to save configuration:

- write
- write memory
- copy running-config startup-config

9. Basic Security Configuration

service password-encryption – Encrypts passwords in configuration.

enable secret – Sets a secure encrypted password for privileged mode.

10. Required Software

- Cisco Packet Tracer
- Terminal Emulator (PuTTY or similar)

11. Procedure

Part A – Device Connection

- Identify console ports on router/switch.
- Connect computer using console/rollover cable.

- Use terminal emulator software to access CLI.

Part B – Packet Tracer Practice

- Open a router in Packet Tracer.
- Access the CLI tab.
- Explore User EXEC, Privileged EXEC, and Global Configuration modes.
- Practice basic commands.

12. Expected Result

Students should be able to access router CLI, navigate between modes, execute basic commands and save configuration.

13. Lab Task

Perform the following in Cisco Packet Tracer:

- Open a router and access CLI.
- Enter privileged mode using enable.
- Enter global configuration mode.
- Configure enable password.
- Enable password encryption using service password-encryption.
- Configure enable secret password.
- Display running and startup configuration.
- Save configuration using copy running-config startup-config.

14. Submission Instructions

- Perform the given lab task.
- Take screenshots of commands and outputs.
- Paste screenshots in a Word file.
- Write 2–4 lines explanation below each screenshot.
- Submit the soft copy to the instructor.